Meeting – Dobril Radoslavov

(Ministry of Defense)

Questions:

1) Може ли да ни даде контакти (основно в Украйна) да ни кажат от първо лице какви са им нуждите и съответно да разберем как да разработим дрона максимално в съответствие с техните нужди

2) Той има ли идея дали има реална нужда от такъв продукт?

3) Използват ли се логистични дронове в момента на бойното поле?

4) Crisis response → при пожари, земетресения, наводнения ...

↳ Гражданска защита е към МВР, но въпреки всичко той има ли някаква информация?

5) С какви дронове разполагат в Украйна? Всичко ли е насочено към камикадзе и surveillance?

6) Какви дронове се разработват в момента в България?

ORGO Meeting 08.02.2026

Open questions:

1) What type of propellers are we going to use? Any CAD files for them?

2) What type of motors are we going to use?

3) Currently, we are drawing the full scale drone. Should we switch to the prototype already?

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1) Go through Abi's systems engineering and give feedback (html file in Discord).

2) 280 x 240 x 300 } Martin's 3D printer
depth x width x heigth

3) Start designing the prototype A/C

ORGO Meeting 15.02.2026

1)

1) Speed is everything nowadays! Look at companies in Asia. They are ahead of us because they are faster.
ORGOHy enables extremely fast deliveries for different companies!

ORGO Meeting 08.03.2026

$\boxed{2 \times 24V}$

↳ Battery clear
↳ Motor clear

~10 000 mAh

2_{prop} →	18 – 22 inch
3_{prop} →	10 – 13 inch

→ точен мотор, който ще използваме

↳ What props exactly?
↳ Control system?

↳ Define V-tail angle and airfoil type

↳ Engine pylons?

1. Mission Definition → **2. Choose the VTOL concept** → **3. Do a first mass estimate** → **4. Size from the hardest mission segment** → **5. Build a rough sizing loop** → **6. Sketch the layout**

1. Mission Definition
- ↳ Payload mass
- ↳ Required range or endurance
- ↳ Cruise speed
- ↳ Hover time
- ↳ Take off / Landing constraints
- ↳ Operating Altitude
- ↳ Wind conditions
- ↳ Safety margins
- ↳ Budget and manufacturing limits

2. Choose the VTOL concept
- ↳ tilt rotor
- ↳ Lift + cruise

3. Do a first mass estimate
- ↳ payload
- ↳ battery
- ↳ propulsion system
- ↳ structure
- ↳ avionics
- ↳ landing gear
- ↳ wiring and margins

4. Size from the hardest mission segment

Usually one of the following:
- ↳ hover power
- ↳ transition control
- ↳ cruise energy
- ↳ wing loading
- ↳ stall speed

22.03.2026

→ крыло

→ V-tail

- Wingcopter (datasheet)
- Joby
- quantum
- skyways
- Zipline (V-tail)

V_{cruise}	→ 24 m/s
V_{max}	→ 42 m/s
V_{stall}	→ 12 m/s

$$\boxed{L = \frac{1}{2}\rho \cdot V^2 \cdot S \cdot C_L} = W$$

$$\Rightarrow S = \frac{2W}{\rho \cdot V_{stall}^2 \cdot C_{L_{max}}}$$

$W = 15 \cdot 9{,}80665 = 147{,}0998\ N$

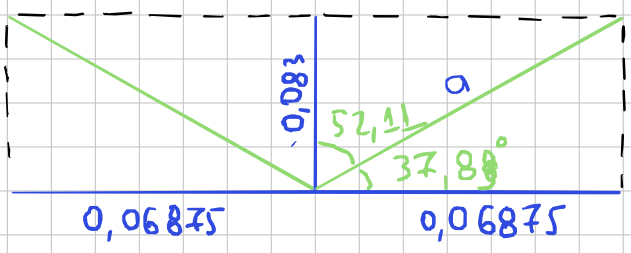
$V_{stall} = 12\ m/s$

$C_{L_{max}}$ → NACA 2412 $\approx 1{,}4$ as first approx.

$Re \approx 500\,000$ at cruise speed and chord 0,3m

$\rho_{600m} \approx 1{,}17$

$$\Rightarrow S = \frac{2 \cdot 15 \cdot 9{,}80665}{1{,}17 \cdot 12^2 \cdot 1{,}4} = 1{,}247\ m$$

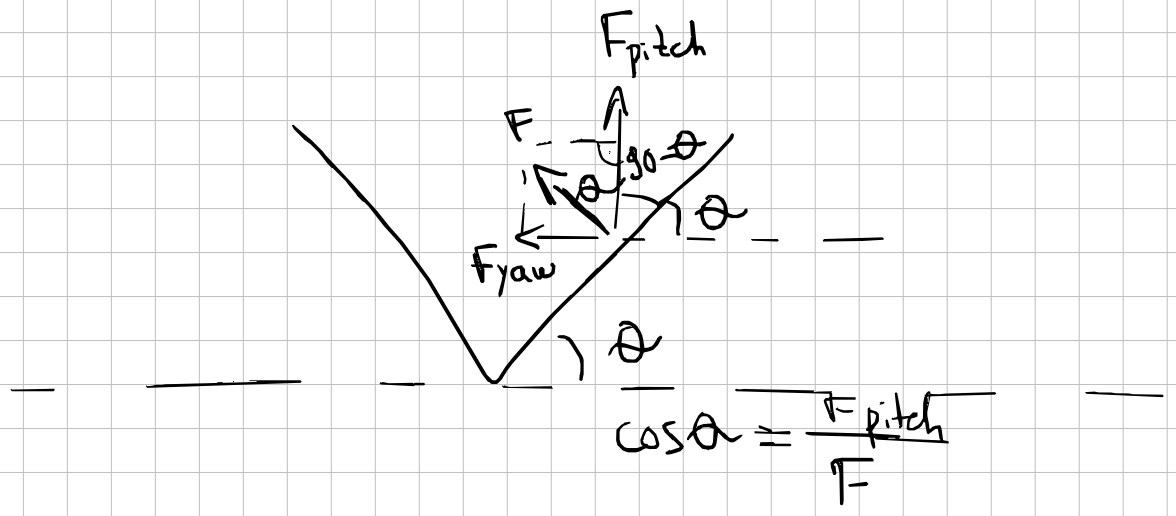


$\cos 52{,}11 = \frac{0{,}083/2}{a}$

$a = \frac{0{,}0415}{\cos 52{,}11} = 0{,}0675$

$\cos 37{,}88 = \frac{0{,}06875}{a}$

$a = \frac{0{,}06875}{\cos 37{,}88} = 0{,}087$



$\cos\theta = \frac{F_{pitch}}{F}$

$F_{pitch} = \cos\theta \cdot F$

$F_{yaw} = \sin\theta \cdot F$

Workflow:

wingspan b_w
wingarea S_w

⊕

choose: C_{HT} / C_{VT}, L_{HT} / L_{VT} → calculate S_{VT} / S_{HT} → calculate V-tail angle θ → calculate V-tail surface area $S_{V\text{-}tail}$

You are controlling only this

To Do: CW 13

1) Add the workflow to the wing and V-tail design worksheet.

2) Find proper components and place them to evaluate the fuselage dimensions

3)

02.04.2026

$$\bar{X}_{CG} - \bar{X}_{AC} \; [\%] \rightarrow 5\% \ldots 10\%$$

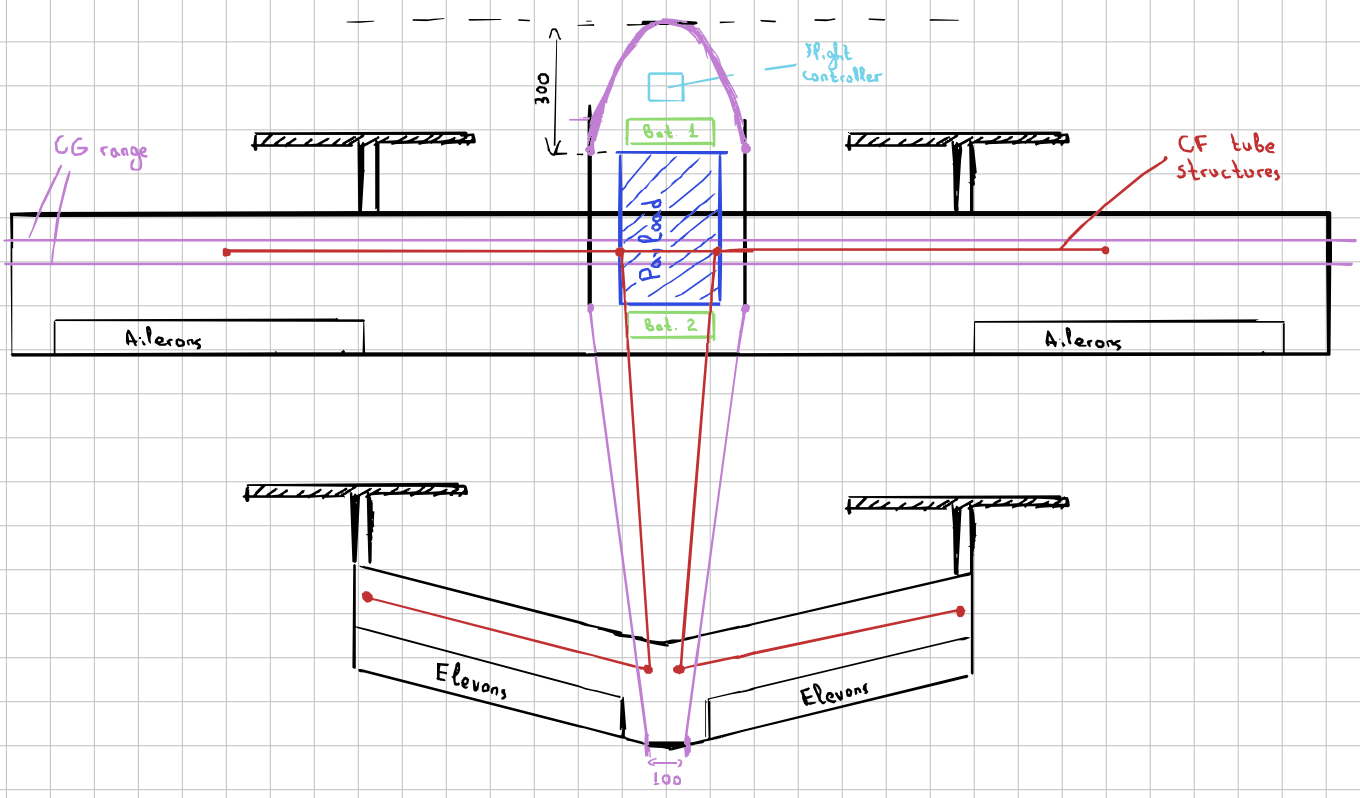
↳ for stable aircraft, the stability measure is negative

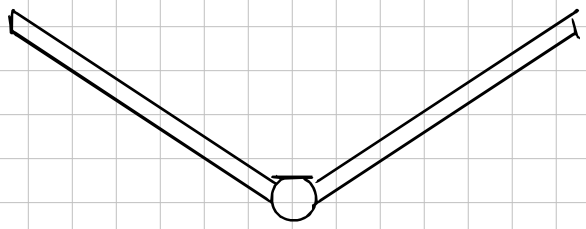
$$\boxed{\bar{X}_{AC} = \frac{C_{M\alpha,w} - C_{M\alpha,fus} + C_{M\alpha,HT}}{C_{L\alpha,w} + C_{L\alpha,HT}}}$$

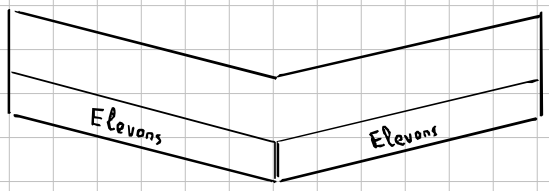
$$\bar{X}_{AC,w} = \frac{X_{0,w} + \left(\overbrace{n_{AC,w}}^{\%} \cdot C_{root,w}\right)}{C_{MAC}}$$

$$C_{L\alpha,w} = \frac{2\pi \cdot AR}{2 + \sqrt{\frac{AR^2 \beta^2}{k^2} \cdot (1 + 0) + 4}}$$

09.04.2026


300
Flight controller
Bat. 1
Payload
Bat. 2
CG range
CF tube structures
Ailerons
Ailerons
Elevons
Elevons
100

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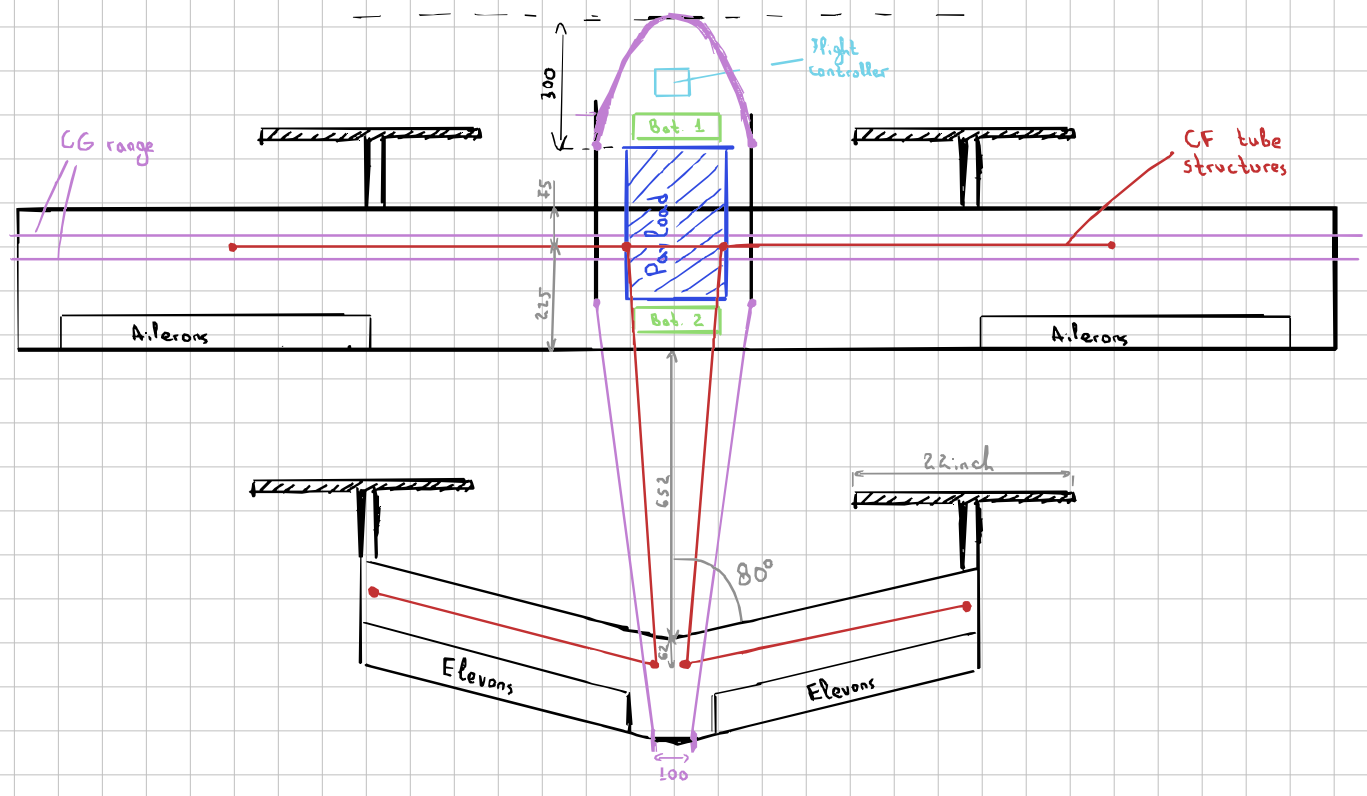
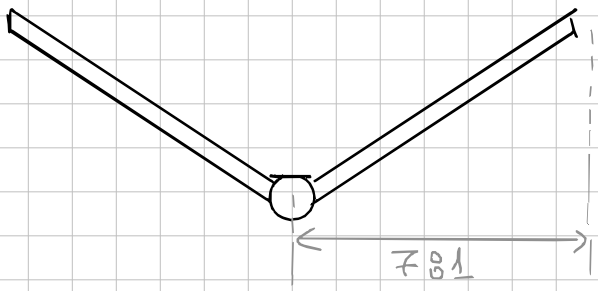
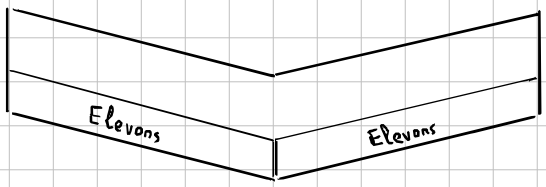
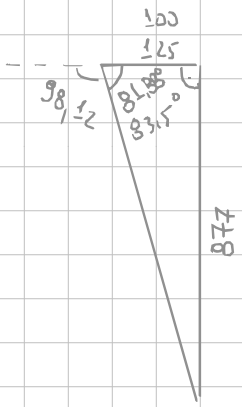


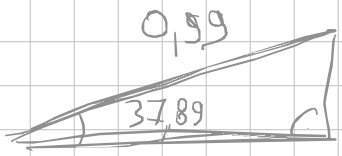






$$tg\alpha = \frac{877}{125} = 0,123$$

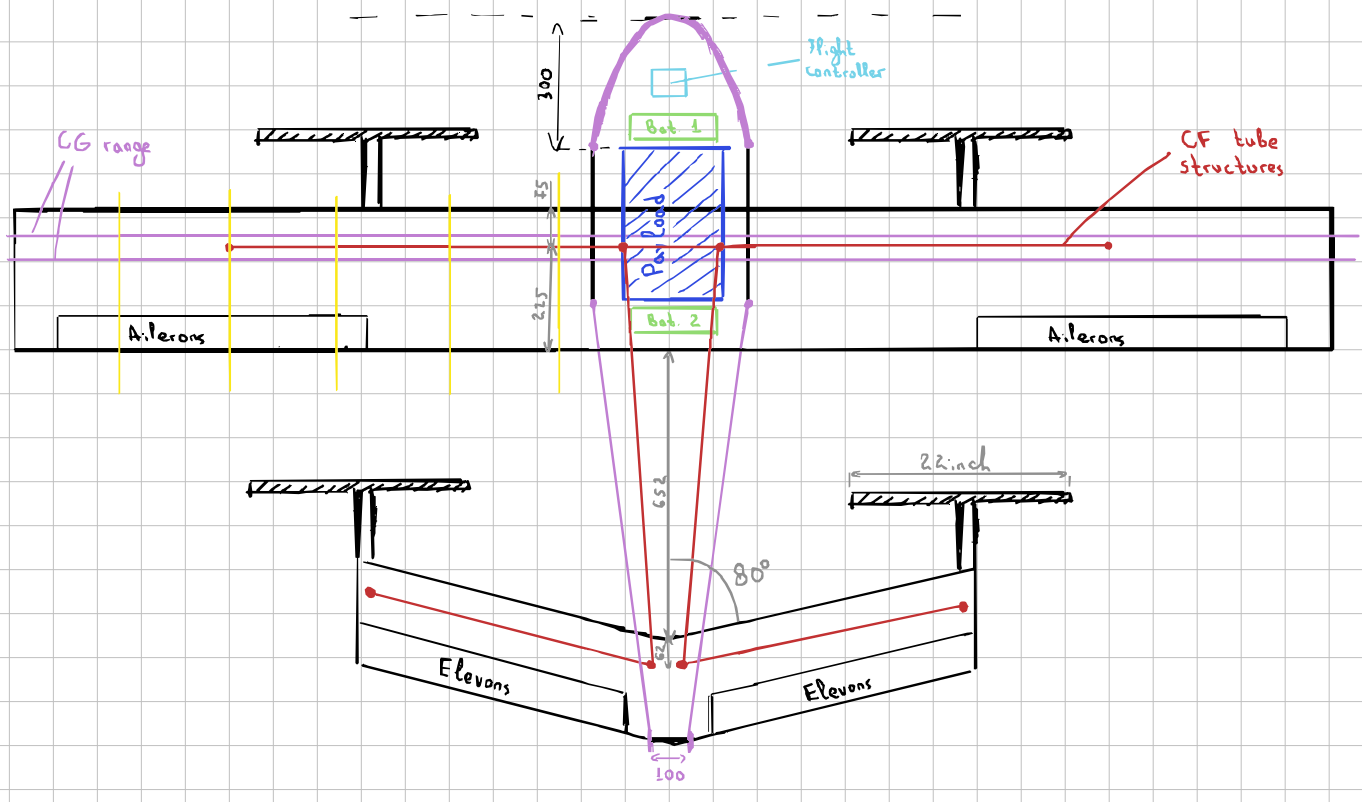
$$tg\alpha = \frac{877}{100} = 0,877$$

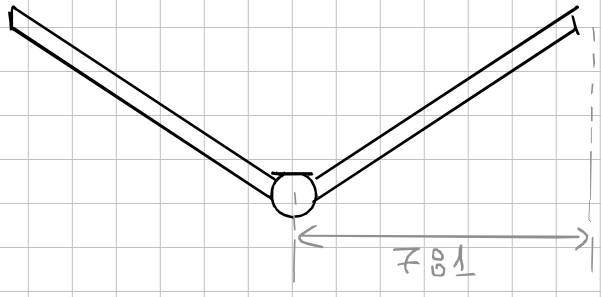


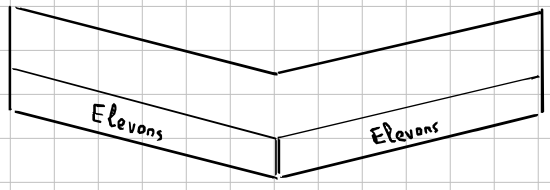
$$x = 0,781$$

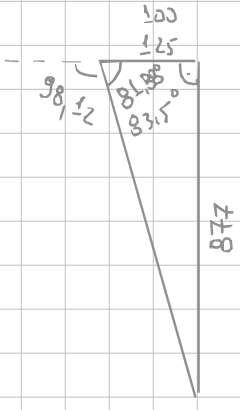
$$\cos 37,89 = \frac{x}{0,99}$$

$$x = \cos 37,89 \cdot 0,99$$









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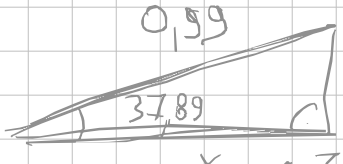
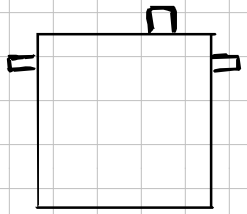


$$x = 0{,}781$$

$$\cos 37{,}89 = \frac{x}{0{,}99}$$

$$x = \cos 37{,}89 \cdot 0{,}99$$

13.04.2026

Tilt Mechanism Concept: